

The Riemann Hypothesis: Cross-Route Synthesis

Part IV of a Five-Part Series

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Abstract

This paper, Part IV of a five-part series [6, 5, 7, 4], collects three classes of material that do not fit the scope of Parts II (Even Dominance) or III (Spectral Routes): the cross-route synthesis comparing all three active branches A, B, Z; the dormant routes and external lines of the wider landscape; and a self-correcting record of hypothesis corrections during the iterative research process.

The cross-route synthesis identifies a shared Galerkin pattern across branches — each conditional reduction involves a fixed-parameter positive structure that must be promoted to a uniform-limit statement via a growing parameter $N(\lambda)$, $d(\lambda)$, or $m(\lambda)$ — and tabulates the characteristic constants ($C^* = 64\pi^2/3$ as the current Branch A boundary-moment benchmark for $\lambda = 5, 7$, $\kappa_\infty \in [1.6, 1.8] \times 10^{-3}$ for Branch B, $g_0(m) \approx 2\pi/\log \gamma_m$ for Branch Z).

Series status (v3.0, 2026-05-26; criticality-lens update 2026-05-31). The series is restructured from a trilogy to a five-part series; the conditional reduction status from v2.2 is unchanged. A 2026 working phase adds Section 3: a *heuristic* criticality lens ($\text{RH} \Leftrightarrow \Lambda = 0$) that strategically re-weights the routes, a sharper localization of the common wall (the analytic continuation across $\Re s = 1$), a computable diagnostic built from the *classical* Laguerre–Pólya/Herglotz criterion, and four honest closures of *concrete constructions* (not routes, and several of them self-corrections of internally proposed constructions). None of these closes RH, refutes a route, or changes the conditional-reduction status.

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1 Introduction

This paper, Part IV of a five-part series, serves as the “miscellaneous and combined” component of the landscape atlas:

Part I [6]: Foundations and Obstructions.

Part II [5]: Even Dominance (Branch A).

Part III [7]: Spectral Routes (Branches B and Z).

Part IV (this paper): Cross-Route Synthesis, dormant routes, external lines, methodical dead ends, hypothesis corrections.

Part V [4]: Conclusio.

The three active branches A (Even Dominance), B (Hadamard Strip / I-1 Normal-Family), and Z (CCM Microcluster) are developed in detail in Parts II and III. This paper identifies the structural patterns they share, surveys the wider landscape of dormant and external routes, and provides a transparent record of hypothesis corrections during the iterative development.

2 Cross-Route Synthesis

2.1 Characteristic Constants

Each active branch produces a characteristic constant or scaling law whose identification anchors the conditional reduction:

Branch	Constant	Scaling law	Theoretical identification
A (Even Dom.)	$C^* = 64\pi^2/3 \approx 210.55$	sLI $\sim C^* \lambda/(NL)$ for the $\lambda = 5, 7$ benchmark; $\lambda = 3$ separate and $\lambda = 11$ open	$16 \int_0^\infty t^2/\sinh^2(t/2) dt$ (reverse-fit candidate; exact collar kernel known)
B (Hadamard)	$\kappa_\infty \in [1.6, 1.8] \times 10^{-3}$	saturates with λ	finite-part residual curvature of Foundation-Lock
Z (CCM)	$g_0(m) \approx 2\pi/\log \gamma_m$	log-decreasing	Montgomery–Odlyzko (classical)

Remark 2.1 (Identification status). The constant $C^* = 64\pi^2/3$ for Branch A is an identification candidate for the $\lambda = 5, 7$ benchmark. The 27 May audit shows that the simple endpoint-kernel factorisation is false; the exact collar kernel is $K(t) = e^t \cosh(t)/(2 \sinh^2 t)$, so a rigorous derivation must pass through the eigenmode asymptotics. The $\lambda = 3$ sweep is recorded as a low-bandwidth outlier, while the single $\lambda = 11, N = 200$ point ($C_{\text{eff}} \approx 168$) remains an open discriminator (Part II [5], Remark on derivation candidate). The constant κ_∞ for Branch B is established empirically in the conservative range $[1.6, 1.8] \times 10^{-3}$ (Part III [7]). The gap $g_0(m)$ for Branch Z rests on classical pair-correlation heuristics.

2.2 The Shared Galerkin Pattern

Despite their different analytical mechanisms, all three branches share a common structural pattern:

1. **Fixed-parameter positive structure:** At each finite (λ, N) (or (λ, m) for Branch Z), the relevant quantity is computable and has the correct sign/bound.

2. **Uniform-limit requirement:** The conditional reduction requires promoting the fixed-parameter statement to a uniform assertion as $\lambda \rightarrow \infty$ with a parameter $(N(\lambda), d(\lambda), m(\lambda))$ growing alongside.
3. **Transient vs. converged regime:** The warm-up parameter is branch-dependent. For the GF1/ C^* branch the current diagnostic scale is $\xi = NL/\lambda$, not N/L : the $\lambda = 5, 7$ rows enter the benchmark window near $\xi \approx 50$, whereas the $\lambda = 11, N = 200$ point remains an open pre-convergence discriminator. Other Galerkin branches still track their own resolution ratios.

This pattern is consistent with a *Bernstein–Galerkin conjecture*: the Galerkin truncation of the Weil quadratic form at dimension N converges to the infinite-dimensional form in the relevant operator topology, and the convergence is uniform in λ after a λ -dependent warm-up phase.

Remark 2.2 (Status of the conjecture). The Bernstein–Galerkin pattern is an empirical observation from 5–7 data points per branch. It has not been formalised as a mathematical statement. Its role is heuristic: it provides a mechanistic explanation for why all three branches have similar structural gaps (fixed-parameter $\checkmark$, uniform-limit open).

2.3 Two Isolated Proof Slots

The cross-route analysis identifies two concrete, isolated proof slots that, if resolved, would close one or more branches:

1. **B-1: Uniform log-curvature control from Foundation-Lock.** If $H_\lambda(\delta) \leq K\delta^2$ with $K \leq 1.8 \times 10^{-3}$ is derived from the Foundation-Lock structure, then Theorem B-1 (Part III [7]) gives $A_\lambda \leq 1.0063$, closing Branch B.
2. **GF1: Eigenmode asymptotics + factor-16 identification.** If the exact collar kernel and the $q_a(k)$ eigenmode asymptotics rigorously produce the factor 16 in $C^* = 16 \cdot 4\pi^2/3$, and the $\lambda = 11$ discriminator does not persist as a separate branch, then Theorem GF1 (Part II [5]) provides a boundary-moment scaling benchmark for the non-exceptional Branch A regime; the $\lambda = 3$ branch remains a conservative guardrail.

These two slots are *independent*: closing one does not affect the other. Each is a well-posed analytical problem amenable to standard tools (complex analysis for B-1; Fourier analysis and spectral theory for GF1).

Remark 2.3 (Both slots are margin-type — a strategic caveat). Both slots are *margin/coercivity* statements: a uniform bound that closes a gap. They remain well-posed open problems, and nothing below refutes them. Section 3 adds a *heuristic* criticality lens: since RH (if true) sits at the critical value $\Lambda = 0$ with no buffer (Section 3.1), exact-structural mechanisms (identity, symmetry, spectral realization) deserve *additional* strategic weight alongside these margin-type slots. This is a re-reading of priorities, not a theorem and not a refutation (see the caveat in Theorem 3.1).

3 The Sharpened Common Wall: A Criticality Lens and an Analytic-Continuation Localization

A 2026 working phase re-examined all branches through one question — *what kind of argument can close the gap?* — and localized the shared obstruction more sharply. Everything in this section is a **strategic lens and a diagnostic localization**, not a new closure of RH and not a refutation of any route. Several items are honest self-corrections of internally proposed constructions. The conditional-reduction status of Parts II/III is unchanged.

3.1 A criticality lens (heuristic, not a theorem)

The de Bruijn–Newman constant Λ satisfies $\Lambda \leq 0 \Leftrightarrow \text{RH}$; Rodgers and Tao (*Ann. of Math.* 2020) proved $\Lambda \geq 0$. Hence $\text{RH} \Leftrightarrow \Lambda = 0$: if RH holds, it sits *exactly* at the critical value, with no margin to spare.

Remark 3.1 (Criticality lens — and its essential caveat). This invites a *strategic* dichotomy of proof mechanisms: *margin-seeking* (a uniform gap / coercivity / smallness bound) versus *exact-structural* (an identity, a symmetry, or a spectral realization that forces the conclusion without slack). The heuristic reading is that a mechanism producing a strict uniform margin would be establishing something with a buffer, which sits uneasily with a critical truth at $\Lambda = 0$. **Caveat (essential, to avoid over-claiming)**. The step “a uniform margin would force $\Lambda < 0$ ” is an *analogy*, not a theorem. The branch-specific margins — the even-dominance gap $\lambda_1^+ < \lambda_1^-$, the Hadamard constant κ , the CCM residual smallness — are *not* shown to equal a de Bruijn–Newman statement; Λ and these quantities are different objects. Consequently the lens does *not* refute the well-posed proof slots of Section 2 (Theorem 2.3); it only suggests weighting exact-structural routes *in addition*. It is a re-reading of priorities, not a result.

Under this lens the routes re-sort by *type*, not by viability: the margin-type branches A (A7/AVG gap), B ((ii-a) coercivity), Z (residual smallness) keep their exact kernels as assets, while the exact-structural lines — M (Bost–Connes $\leftrightarrow$ Meyer, with the KMS phase transition at $\beta = 1$ playing the role of built-in criticality) and the global adelic trace route (Connes–Consani scaling site / $\mathbb{F}_1$, where the only missing datum is the Hodge-index positivity of an otherwise-available Lorentzian form) — receive additional strategic weight. The empirical signature is consistent: the even-dominance margin grows like a power law $\sim \lambda^{0.72}$ (a criticality signature, not a buffer). This re-weighting is recorded in the project decision log; it does not alter any route’s formal status.

3.2 The common wall: analytic continuation across $\Re s = 1$

Independently of the lens, the correctly-typed routes share a single concrete endpoint. Three lines of work — the Möbius prime-hub, the Wiener–Hopf factorization, and the Sonin/Toeplitz (Connes–Consani) construction — are each *locally/finitely* controllable but break at *exactly* the same place: the global limit, where the local data must assemble into ζ . The unified description is the Weyl–Titchmarsh function M_∞ of the unbounded limit operator; the missing datum is uniform control of that limit from local data *without* importing ζ .

Remark 3.2 (The wall is a classical fact, here sharply localized). A finite Euler product $\prod_{p \in P} (1 - p^{-s})^{-1}$ has, after the substitution $z = -i(s - \frac{1}{2})$, a pole comb in the upper half-plane at $z_{p,k} = 2\pi k / \log p + i/2$ (on $\Im z = \frac{1}{2}$), with a common pole at $z = i/2$ of order $|P|$; hence its logarithmic derivative is not a Herglotz function. The comb is removed only by ζ 's analytic continuation ($\zeta(0)$ finite; the pole at $s = 1$). This is **Euler-product folklore** (the product converges only for $\Re s > 1$). It is not a new obstruction: it is a *precise localization* of the known wall — the missing datum is the analytic continuation across $\Re s = 1$, i.e. a global/functional-equation phenomenon that finite arithmetic-plus-archimedean data structurally do not supply.

3.3 A computable diagnostic from the classical Laguerre–Pólya/Herglotz criterion

Write $\Xi(z) = \xi(\frac{1}{2} + iz)$ and $M_\Xi = -\Xi'/\Xi$. The chain

$$\text{RH} \iff \Xi \in \text{Laguerre-Pólya} \iff M_\Xi \text{ Herglotz} \iff \frac{(M_\Xi(z) - \overline{M_\Xi(w)})}{z - \bar{w}} \succeq 0$$

is **classical** (Hermite–Biehler; the Laguerre–Pólya characterization of real-rooted entire functions; the positivity $\Re \xi'/\xi > 0$ for $\Re s > \frac{1}{2}$ is Voros' criterion). *We claim no novelty for the criterion itself.* The contribution is a *computable, non-circular test protocol* that turns this classical equivalence into a diagnostic instrument: an Euler-anchor giving M_Ξ from ξ'/ξ in the convergent region $\Re w > 1$ (verified numerically to $\sim 10^{-31}$), a ladder of gates (normalization; tail-tightness; a Stieltjes identity; an off-axis Poisson tomography resolving individual residues), and a detector lemma by which an off-line zero forces a negative 1×1 Pick minor just below the migrated pole. The instrument is empirically validated: it detects off-line zeros and discriminates the true zero spectrum from a smooth density-matched control.

Remark 3.3 (What the diagnostic established on a genuine operator). An independent, non-circular discretization of the Connes–Moscovici prolate operator (arXiv:2112.05500) was run through the full instrument. It reproduces Riemann's zero-counting function $N(E)$ to $\sim 3\%$ in the UV — the *archimedean half* (the smooth density, leading term $\frac{1}{2} \log(y/2\pi)$), an independent non-circular confirmation of the CCM density claim. But its poles sit $O(1)$ off the zeros, so the arithmetic fine-structure (the *positions*) is not captured; and the relative-determinant / Krein construction does *not* remove the Euler pole comb, because the prolate background is itself Herglotz (comb-free), leaving nothing to cancel. Both findings re-confirm Theorem 3.2: density yes, positions no; the wall is the analytic continuation. The genuine positive by-product — that the determinant channel $-D'/D$ (residues = multiplicities), not mere eigenvalue convergence, is the object that must converge ($\det_{\text{reg}}(A - z) \rightarrow C \cdot \Xi$) — sharpens the CCM frontier and is recorded among the independent results (Section 7).

3.4 Four closures of concrete constructions (not routes)

The phase produced four genuine negative results, each over a *specific constructed object or technique*. None closes a whole route; each re-types the route or names a successor.

1. **Osterwalder–Schrader reflection, natural form.** The reflection block $Q_+(x) = \int \int_0^\infty x(t) K(t+s) x(s)$ is indefinite, because the prime distributions $\delta(t+s - \log p)$

in the Weil kernel meet the half-axis $t + s = \log p$; the Toeplitz alternative $K(t - s)$ collapses to global Weil positivity = RH. The OS route in its most natural form (inversion reflection on $\{|g| \geq 1\}$) is closed-negative; another half-space choice would be needed.

2. **Möbius $\mathbb{Z}_2$ prime hub, naïve L_s^- .** An obstruction theorem: any continuation of the divergent prime traces $\text{Tr } T_s = P(s)$, $\text{Tr } T_s^2 = P(2s)$ down to $\Re s = \frac{1}{2}$ imports ζ and its zeros via $P(s) = \sum_n \mu(n)/n \cdot \log \zeta(ns)$ — hence circular. (Project-internal; not a literature-verified standard.)
3. **Toeplitz-symbol sub-method (Branch A).** $A_{\sin}$ is not Toeplitz (m -dependent oscillating diagonals $\Rightarrow$ no well-defined symbol $f(\theta)$); a structural defect of the constructed object, not merely “the method equals the open problem”. Successor: the finite Sylvester-inertia bound (done) plus GLT/outlier control.
4. **Relative-determinant / Krein bypass.** The relative determinant $-\partial_z \log[\det(A - z)/\det(A^0 - z)]$ does *not* remove the Euler pole comb: if A^0 is the prolate background it is itself comb-free (nothing to cancel, and $\det(\text{Euler} \cdot \text{prolate})/\det(\text{prolate}) = \det(\text{Euler})$ keeps the comb); the only comb-free alternative needs a self-adjoint prime factor, which is circular (= RH). This is an honest self-correction of an internally proposed “right technology”; the proposed-then-refuted arc is recorded in Section 8.

Crucially, the *determinant channel* $-D'/D$ as a *sharpening* of the CCM frontier remains valid (Section 7); only the *relative determinant* as a *bypass* of the continuation wall is refuted. The two must not be conflated.

4 Dormant Routes

A small number of constructed routes are dormant rather than dead: they are explicit reductions or partial constructions that have not been refuted, but currently lack active development. The internal register (AKTIONSPLAN.md) keeps the full inventory; the four below are summarised here because each has a direct bridge to one of the active branches.

- **D-CCM (Dirichlet–CCM operator $D_{\log}^{(\lambda, N, \chi)}$).** A direct χ -twist of the CCM rank- N perturbation construction. Addresses GRH, not RH directly — the trivial character χ_0 (Riemann case) is in this scaling regime not practically testable. Companion of Branch Z.
- **D-Twist (χ -twisted defect norm).** Diagnostic blueprint with completed Pilot v2 and Milestone 2_χ work; reformulated as λ -asymptotic statement. First tests at χ_{21}, χ_{33} ; Riemann unreachable in this scale.
- **D-PW (Paley–Wiener sector difference $\psi_\chi = \phi_\chi^+ - \phi_\chi^-$).** Classification of the five Paley–Wiener steps complete: S1, S2, S4 mechanical; S3, S5 character-specific. Continuation of the published Dirichlet Character Atlas v2 (Zenodo 20241612).
- **P-M (Möbius-graded $\mathbb{Z}_2$ prime hub).** Conditional lemma proved for finite prime set (Möbius identity exact); full prime set converges only for $\Re(s) > 1$; $\mathbb{Z}_2$ -sector statement for non-trivial zeros open. Structurally appealing alternative

interpretation of the Prime Orbit Axiom (Part I, Definition 2.1). A 2026 obstruction theorem sharpens this (Section 3.4): no non-circular L_s^- exists, since the divergent prime traces continue to $\Re s = \frac{1}{2}$ only by importing ζ and its zeros. This re-types P-M as the *prime half* of a P-M+M fusion — it provably lacks the archimedean-modular half ($s \leftrightarrow 1 - s$) that route M supplies (project-internal).

The remaining dormant entries (P-A, P-C, P-B, M, R, H) are not listed individually here; see `AKTIONSPLAN.md` for the full inventory. Two are *strategically elevated* by the criticality lens of Section 3.1: **M** (Bost–Connes $\leftrightarrow$ Meyer) and the global adelic trace route **P-A** (Connes–Consani scaling site) are of the exact-structural type and now receive additional strategic weight; **P-C** is co-resolved by an active branch; **P-B** (Berry–Keating) is of the right type but unconstructed (self-adjointness is the missing Hilbert–Pólya theorem); **R**, **H** are framework/expository.

5 External Lines

The landscape sits inside a wider research ecosystem. The following lines are cited as anchors; only the directly proof-relevant items appear here.

- **Connes 2026 Mainline** [1] (arXiv:2602.04022). Re-frames the Riemann zeros as the infrared spectrum of a self-adjoint operator and identifies the key open hypothesis MS2 (§6.6(b)). Branches A, B, Z all condition on this line.
- **Connes–van Suijlekom 2025** [3] (arXiv:2511.23257). Theorem 5.6: 1-dim even kernel of the truncated Weil form implies real zeros of $\hat{\eta}$. This is A1 = B1 in the present landscape.
- **Connes–Consani–Moscovici 2025** [2] (arXiv:2511.22755). Foundation of the rank-one perturbation $D_{\log}^{(\lambda, N)}$. Anchors Branch B’s Foundation-Lock and Branch Z’s microcluster construction.
- **Groskin 2026** (arXiv:2605.20224). Independent implementation of the CCM Galerkin matrix at sixteen cutoffs $c = 13 \dots 67, 100$ at high precision (Aitken extrapolation, $\sim 10^{-334}$ depth). Acts as external validator for branches B and Z.
- **Śliwiński 2026** (arXiv:2601.12133). Counter-regime: in the $\lambda = N$ configuration with low precision, proves an inverse-logarithmic dissonance lower bound $\epsilon(\lambda, N) \geq 1/(4 \ln \lambda)$. Does not bound the fixed- λ , $N \gg \lambda$, high-precision setup of this series; stands as a warning against naïve $\lambda = N$ limits.
- **Watson 2026** (arXiv:2602.01248), “The Riemann Ξ -function from primitive Markovian cycles I.” Constructs a Laguerre–Pólya function Ψ from finite reversible Markov dynamics, “entirely Archimedean, using no Euler products, primes, or arithmetic spectral input”; not an RH proof — it isolates “a single remaining analytic problem relating Ψ to $\Xi(2 \cdot)$.” An independent triangulation of the common wall: from the archimedean side it reaches the same *identification* step (relating the constructed object to Ξ) that the routes of this series reach from the arithmetic side (Section 3.2).

- **Franchi Viceré 2026** (Zenodo 19546495). Claims unconditional RH proof via apeirohedron / semi-local spectral descent. Surveyed and found not viable: the stabilisation argument confuses pointwise with strip-uniform convergence; the Apeirohedron limit step is not rigorous; boundary conditions do not replace the even-sector / MS1 input. See `AKTIONSPLAN.md` §Klasse 8.

6 Systematically Explored Alternatives

Eleven alternative strategies (BI-1 through BI-11) were systematically explored during development. This exhaustive search is itself a meta-result: the solution space is mapped.

Approach	Core idea	Result	Status
BI-1: $\det_2$ / Carleman	Extend to Hilbert–Schmidt class	No positivity substitute	Refuted
BI-2: RG fixed point	Weil positivity under scaling	Not formalizable	Discarded
BI-3: Universal property	Physical plausibility $\Rightarrow$ RH	Too abstract	Discarded
BI-5: Lifting programs	Weil $\rightarrow$ Grothendieck $\rightarrow$ RMT	All open except $\mathbb{F}_q$	Not usable
BI-6: Implication ring	$Li \leftrightarrow Weil \leftrightarrow NB \leftrightarrow BD$	Ring does not exist	Refuted
BI-7: Pöschl–Teller scattering	Gamma structure analog to ξ	No off-line stability	Negative
BI-8: Second-order stability	RH as variational problem	Architecture correct	Realized
BI-9: Soliton models	Topological stability	Coulomb gas unstable	Refuted
BI-10: Building blocks	No-Coordination, shift structure	Successfully reused	Integrated
BI-11: Connes 2026	Even dominance via Q_W	Breakthrough	Core path

The chosen path (BI-8 + BI-10 + BI-11) combines the variational architecture, the arithmetic building blocks, and Connes’ spectral framework. Routes A and B (total positivity, commuting operator) were ruling-outs that are structural discoveries: even dominance holds *despite* their absence, which clarifies that the mechanism is genuinely arithmetic-statistical rather than algebraic (Part II [5]).

7 Independent Results

Several results from this program are of independent interest, regardless of the main proof’s status:

Result	Context	Significance	Status
<i>Positive results</i>			
Shift Parity Lemma	Arithmetic trigonometry	Every prime favors even eigenfunctions	Proved
No-Coordination Lemma	Multiplicative independence	Arithmetic substitute for entropy term	Proved
Leading-Mode Cancellation ($c = 2 + \sqrt{2}$)	Resolvent analysis	Universal pairwise cancellation constant	Proved
Euler–Maclaurin: $\rho^{\text{EM}} < 0$	PNT asymptotics	Global stability anchor	Proved
33 finite-range CAP diagnostics	Interval arithmetic	Conditional bridge	Evidence
Concentration lemma	Even-dominance image measure	No fixed prime set controls a positive share of prime mass	Proved
Finite Sylvester inertia ($\nu_- = 0$)	Interval arithmetic, $\lambda = 100$	Finite even dominance rigorous, no symbol needed	Verified
Residue-calibration channel	Weyl-function diagnostic	Eigenvalue conv. $\neq$ Weyl conv.; channel is $-D'/D$, $\det_{\text{reg}} \rightarrow C\Xi$	Diagnostic
<i>Negative results</i>			
Obstructions K1–K4	Dead-end analysis	Maps impossible strategies	Documented
No absolute PF_∞ for A_λ	Prime shift multiplier	$M(\xi)$ negative on $\sim 49\%$ of domain	Proved
No universal commuting operator	Sturm–Liouville search	Kernel of $[T, D_N(r)]$ is $\{c \cdot I\}$	Proved
<i>Structural insights</i>			
Inverted-Hessian structure	YM/TU/RH comparison	RH is a maximum, not minimum	Confirmed
Cross-disciplinary catalog	Regularization	11 fields share trace-class obstruction	Documented

The negative results protect future researchers from repeating dead-end strategies. The Shift Parity Lemma, the obstruction analysis (K1–K4), and the Mellin decay barrier are publishable independently.

8 Hypothesis Corrections in the Iterative Process

The iterative development produced several hypothesis corrections that are documented here as a record of honest research practice (Sammler-Disziplin).

8.1 GF1 Correction Chain (Iterations 15–29)

Iter	Hypothesis	Status
15	$C_m \approx 8$ universal	falsified (boundary effect)
18	$C_m(\lambda = 5) \approx 4.3$	falsified (no plateau)
19	$\text{sLI} \sim 1/N$	confirmed, but without λ -scaling
20	$N \cdot \text{sLI} \approx 329$ universal	falsified
27	$\text{sLI} = \mathbf{C}^* \lambda / (\mathbf{NL}) + \mathbf{o}(\cdot)$	confirmed for $\lambda = 5, 7$, $C^* = 64\pi^2/3$
28	$\lambda = 3$ all- λ check	low-bandwidth outlier; two-regime guardrail
29	$\lambda = 11$ discriminator and endpoint-kernel audit	$C_{\text{eff}} \approx 168$ at $N = 200$; simple endpoint factorisation falsified

8.2 B-1 Corrections and c_{Ξ} RH-Neutral Form

The Hadamard constant c_{Ξ} was initially written in the RH-specialised form $\sum_{k \geq 1} 1/\gamma_k^2$ (sum over imaginary parts of zeros on the critical line). An external audit corrected this to the RH-neutral form $c_{\Xi} = -\frac{1}{2} \sum_{\rho} 1/(\rho - \frac{1}{2})^2$ (sum over all non-trivial zeros). This is a Sammler-Disziplin correction: the RH-neutral form is mathematically more general and does not assume what is being proved.

8.3 The Relative-Determinant Bypass: Proposed and Refuted (2026)

During the 2026 phase, the relative determinant (Krein spectral shift) $-\partial_z \log[\det(A - z)/\det(A^0 - z)]$ was proposed internally as the “right technology” to remove the upper-half-plane Euler pole comb and so bypass the continuation wall (Theorem 3.2). A subsequent test on the genuine Connes–Moscovici prolate background refuted this as a bypass: the prolate A^0 is itself Herglotz (comb-free), so there is nothing to cancel ($\det(\text{Euler} \cdot \text{prolate})/\det(\text{prolate}) = \det(\text{Euler})$ retains the comb), and the only comb-free alternative needs a self-adjoint prime factor, which is circular (= RH). The proposal is withdrawn as a bypass. The *separate* finding — that the determinant channel $-D'/D$, not mere eigenvalue convergence, is the correct object to converge ($\det_{\text{reg}}(A - z) \rightarrow C \Xi$) — remains valid (Section 7); the two must not be conflated.

A second correction: an internal phase lemma $\Theta_{\sigma} = \xi(\sigma - iz)/\xi(\sigma + iz)$ was sign-reversed (its poles come from the denominator, so under RH it *has* poles); the correct object is the reciprocal phase $\hat{\Theta}_{\sigma} = \xi(\sigma + iz)/\xi(\sigma - iz)$, with $\text{RH} \Leftrightarrow \hat{\Theta}_{\sigma}$ pole-free in the upper half-plane for all $\sigma > \frac{1}{2}$.

8.4 Value of the Correction Record

These corrections demonstrate that the research process is self-correcting. Each correction sharpened the theoretical framework: the GF1 chain moved from a naïve constant ($C_m \approx 8$) through several intermediate forms to the $\lambda = 5, 7$ boundary-moment benchmark $\text{sLI} \sim C^* \lambda / (NL)$; the $\lambda = 3$ sweep and the open $\lambda = 11$ discriminator prevent an all- λ theorem reading, and the c_{Ξ} correction removed a circular RH-dependence from a key constant.

9 Summary

Part IV contributes the cross-route perspective: the three active branches share a Galerkin pattern (fixed-parameter positive, uniform-limit open), and two isolated proof slots (B-1 log-curvature, GF1 factor-16 identification) are the concrete analytical targets, which remain well-posed open problems. Section 3 then adds a 2026 *heuristic* criticality lens ($\text{RH} \Leftrightarrow \Lambda = 0$) that re-weights the routes toward exact-structural mechanisms (without refuting the margin-type slots), localizes the common wall as the analytic continuation across $\Re s = 1$, records a computable Laguerre–Pólya/Herglotz diagnostic (a classical criterion turned into a test instrument), and reports four closures of *concrete constructions* — the OS reflection in natural form, the naïve Möbius L_s^- , the Toeplitz-symbol sub-method, and the relative-determinant bypass — none of which closes a route or RH. The dormant and external routes are catalogued for completeness; the hypothesis correction record provides transparency. The condensed atlas and final assessment follow in Part V [4].

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AI Disclosure

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